

Meal Prep Four Containers SOP

One episode fills four meal containers with the same three components, seats a lid on each one, and builds them into one stack.

The order never changes: container, component 1, component 2, component 3, lid, stack. That runs once per container, four times. Component 1 always goes in first and component 3 always goes in last, a container is never carried off the fill spot without a seated lid, and the stack never receives an open container.

No gripper ever touches a component. Component 1 moves by scoop and every piece of component 2 and component 3 moves by tongs. Only one tool is held at a time, and it goes back to the tool rest before the other is taken.

The right gripper does the portioning: it takes the scoop, takes the tongs, fills the compartments, and seats the lids. The left gripper supports: it moves each container to the fill spot, holds it down while the components go in, and carries the lidded container to the stack. Neither gripper carries anything to the other side of the table.

The left gripper's hold on the container stays on from the fill spot to the stack. It sits low on the outside wall near the base, clear of the container rim, so it holds the container through the portioning and through the lid press and the lid skirt still comes down over the rim all the way around. Nothing ever comes down onto an unheld container, and the same closed grasp becomes the lift that carries the container to the stack, so the gripper opens only once the container is down on it.

The one exception to the arm split is the **steady assist**, used in Step 1 and nowhere else. An empty container tips or slides when the left gripper closes on it in the container row, so the right gripper may reach in empty-handed and hold it still while the left gripper takes its grasp. The right gripper never lifts the container, never moves it, and releases before the left gripper lifts. It never holds a container at the fill spot, while a component goes in, or while the lid is seated. The lidded container needs no assist, because the left gripper is still holding it when the carry to the stack begins.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the three source bins across the back right, the lid rest right of the fill spot, the tool rest at the front right, the fill spot at the center, the container row at the front left, and the stack spot left of the fill spot.
3. The fill spot is visible from above, so every scoop and every piece can be seen landing in its compartment and anything that misses can be seen on the table.
4. Both arms are at home with grippers open.
5. The table is clean, dry, and bare apart from the bins, the containers, the lids, and the tools.
6. The right arm reaches the three source bins, the fill spot, the lid rest, and the tool rest. The left arm reaches the container row, the fill spot, and the stack spot.
7. The right arm reaches every container in the container row with its gripper empty, for the steady assist in Step 1. Check the reach at the leftmost container, the farthest one. If the right arm cannot reach it, slide the whole row right until it can while the left arm still reaches every container, and confirm the row is still clear of the carry path from the fill spot to the stack spot. If the row cannot sit inside both arms' reach, run the episode without the assist: the left gripper picks every container in the row on its own, which is correct and is not a violation. The assist is the only reach the right arm makes to the left, so nothing else in the episode depends on it.
8. The left arm reaches the stack spot with four lidded containers standing on it, so the fourth container is set on the stack without the arm extending to a joint limit.

Materials checklist

1. Three open **source bins** stand in a row across the back right: **bin 1** nearest the center, then **bin 2**, then **bin 3** at the far right. Each bin holds one component and only that component.
2. Bin 1 holds a loose component that levels off flush with the scoop edge, with material to spare past eight scoops.
3. Bin 2 holds separate pieces, each small enough to sit inside the left compartment, more than sixteen in the bin.
4. Bin 3 holds separate pieces, each small enough to lie flat in the right compartment, none stuck together, more than twelve in the bin.
5. Four clean dry **containers** stand open in the container row at the front left, in one row running left to right, each with its base flat and its **large compartment** nearest the front edge.
6. Four matching **lids** are stacked at the lid rest right of the fill spot, inner face down, with no lid interior touching the table.
7. One **scoop** and one pair of **tongs**, both clean and dry, rest in their own spots at the tool rest at the front right, handles clear of each other.

8. The fill spot at the center of the table and the stack spot left of it are clear.

Workspace layout

- **Source bins:** back right, bin 1 nearest the center, then bin 2, then bin 3.
- **Lid rest:** right of the fill spot, the four lids stacked inner face down.
- **Tool rest:** front right, the scoop in one spot and the tongs in the other.
- **Fill spot:** the center of the table, where the one open container is portioned and lidded.
- **Container row:** front left, the four empty open containers in one row.
- **Stack spot:** left of the fill spot, where the stack of lidded containers builds.

Everything the right arm works on is staged on the right or in the center. Everything the left arm carries moves along the left. No carried thing crosses the midline. The right gripper reaches left of the midline for one move only, the empty-handed steady assist on a container in the container row, and it carries nothing when it does.

Vocabulary

- **Arm assignment:** the right gripper takes the tools, fills every compartment, and seats every lid. The left gripper moves each container to the fill spot, holds it there through the portioning and the lid press, and carries it lidded to the stack spot on that same grasp. Neither gripper takes the other's zones, apart from the steady assist.
- **Steady assist:** the right gripper, empty, closed on the outside wall of a container on its right side near the base, holding it still while the left gripper closes on the opposite wall. It is used in one place and nowhere else: on an empty container in the container row in Step 1. The container stays flat on the table throughout. The right gripper opens and draws clear as soon as the left gripper is closed, before the container lifts. It is a hold only: the right gripper never lifts, carries, slides, turns, or rights a container, and never holds one at the fill spot, while a component goes in, or while the lid is seated. It is used at the operator's discretion, on any container in the row or none of them.
- **Component 1:** the loose component in bin 1, moved by scoop. **Component 2** is the pieces in bin 2 and **component 3** is the pieces in bin 3, both moved by tongs. A component only ever enters its own compartment.
- **Container:** one meal container divided into three compartments. The **large compartment** runs along the side nearest the front edge. Behind it sit the **left compartment** and the **right compartment**, split by a **divider**. The **container rim** is the raised outer edge the lid seats on.
- **Portion:** what one container receives. Two level scoops of component 1 in the large compartment, four pieces of component 2 in the left compartment, three pieces of component 3 in the right compartment.

- **Level scoop:** the scoop filled and drawn along the bin rim so component 1 sits flush with the scoop edge and is not heaped above it.
- **Scoop grip:** the right gripper closed on the middle of the scoop handle, never on the scoop bowl.
- **Tongs grip:** the right gripper closed across the middle of the tongs arms. Closing the gripper further closes the tongs on a piece. Easing the gripper back opens the tongs and releases the piece without dropping the tongs.
- **Tool home:** the scoop's spot and the tongs' spot at the tool rest. A tool is in one gripper or at its tool home, never on the table and never in a bin.
- **Container hold:** the left gripper closed on the outside wall of the container at the fill spot, on its left side near the base, keeping the base flat on the table. It is taken low on the wall, well below the container rim, so the **lid edge** comes down over the rim without meeting the gripper. It runs unbroken from the moment the container lands at the fill spot until the lid is seated, and the gripper does not open at the end of it: the same closed grasp becomes the lift that starts the carry to the stack in Step 6. The container stays flat on the table for the whole of the hold.
- **Lid:** one lid per container. Its **lid edge** is the outer skirt and the only part a gripper holds. Its **lid interior** is the face that meets the container rim. Its **lid top** is the flat outer face, which no gripper presses on.
- **Seated lid:** the lid lying flat and square on the container rim, its edge down all the way around, with no corner standing proud and no gap along any side.
- **Stack:** the lidded containers at the stack spot, each squared on the one below with their edges in line and every large compartment nearest the front edge.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Release point:** over a compartment, over the fill spot, over the stack spot, over a tool home, or over the lid rest. A gripper opens nowhere else.
- **Debris:** any component outside a source bin, a tool, or a compartment.

Steps

Steps 1 to 6 are one container and run four times, taking the containers in order from the left end of the container row. Step 7 runs once, after the fourth container is stacked.

Step 1: Move a container to the fill spot

Goal: one empty open container flat at the fill spot, facing front, held by the left gripper.

- If the next **container** in the container row needs steadying, the **right gripper**, empty, comes down onto its right-side outside wall near the base and closes just enough to hold it still on

the table. This is the **steady assist**. The right gripper touches a container here and nowhere else.

- The **left gripper** closes on the outside wall of the next **container** in the container row, lifts it straight up, carries it level to the **fill spot**, and lowers it until its base is flat and centered with the **large compartment** nearest the front edge.
- If the steady assist is in use, the **right gripper** opens and draws clear of the container row once the left gripper is closed on the container and before the container leaves the table. The right gripper does not travel with the container.
- Without releasing, the left gripper settles into the **container hold**, low on the wall and well below the **container rim**, and keeps it through the portioning in Steps 2, 3, and 4 and through the lid press in Step 5. It does not open again until the container is down on the stack in Step 6.

Take one container. Never lift two in one grasp and never drag a container across the table.

The steady assist holds, it does not move. The right gripper stays empty for it, so it never comes in with the scoop, the tongs, or a lid in hand. If the container ends up shifted out of the row or tipped over, the left gripper is the one that rights it and sets it back. The container never lifts off the table with both grippers on it.

Check: one empty open container sits flat and centered at the fill spot with the large compartment nearest the front edge, the left gripper is on its left wall near the base, the right gripper is empty and clear of the container row, and all three compartments are clear.

Step 2: Portion component 1

Goal: two level scoops of component 1 in the large compartment and the scoop back at its tool home.

2.1 Take the scoop

- The **right gripper** closes on the middle of the **scoop** handle at the tool rest in a **scoop grip** and lifts it clear.

2.2 Fill and empty the scoop

- Push the scoop bowl into **bin 1**, fill it, and draw its edge along the bin rim on the way out so the component sits flush with the scoop edge as a **level scoop**.
- If the scoop comes up heaped, lower it back into bin 1 and draw its edge along the bin rim again. Never level a scoop over the container.
- Carry the scoop level to the fill spot, hold it over the **large compartment**, and tilt it until the component falls in, then bring it back level.
- Repeat once so two level scoops go into the large compartment.

2.3 Return the scoop

- The **right gripper** carries the scoop back to its **tool home** at the tool rest, lowers it, and releases.

Check: the large compartment holds two level scoops, none of component 1 is in the left compartment or the right compartment, nothing stands above the container rim, and the scoop is at its tool home.

Step 3: Portion component 2

Goal: four pieces of component 2 in the left compartment.

- The **right gripper** closes across the middle of the **tongs** arms at the tool rest in a **tongs grip** and lifts them clear.
- Lower the tongs into **bin 2**, close the gripper further so the tongs take one piece, and lift it clear of the bin.
- Carry it level to the fill spot, lower it into the **left compartment**, and ease the gripper back until the tongs release it.
- Repeat until four pieces sit in the left compartment.
- Keep the tongs in the gripper for Step 4.

One piece per pick. If a piece drops on the table, leave it where it fell and take a fresh one from bin 2. Anything that has touched the table never goes into a container.

Check: four pieces of component 2 sit in the left compartment, none on the divider, in another compartment, on the container rim, or above it.

Step 4: Portion component 3

Goal: three pieces of component 3 in the right compartment and the tongs back at their tool home.

- With the tongs still in the **right gripper**, lower them into **bin 3**, close the gripper further so the tongs take one piece, and lift it clear of the bin.
- Carry it level to the fill spot, lay it flat in the **right compartment**, and ease the gripper back until the tongs release it.
- Repeat until three pieces lie flat in the right compartment.
- The **right gripper** then carries the tongs back to their **tool home** at the tool rest, lowers them, and releases.

One piece per pick. If a piece drops on the table, leave it where it fell and take a fresh one from bin 3.

Check: three pieces of component 3 lie flat in the right compartment, none on the divider, in another compartment, or above the container rim, the tongs are at their tool home, and the left

gripper is still in the container hold.

Step 5: Seat the lid

Goal: a seated lid on the full container, with the left gripper still holding the container.

- Confirm no piece stands above the container rim. If one does, take the tongs again, lower it into its compartment, and return the tongs before going on.
- The **left gripper** keeps the **container hold** through the whole of this step. It stays low on the wall near the base, well below the **container rim**, so the lid comes down clear of it.
- The **right gripper** closes on the **lid edge** of the top lid at the lid rest, lifts it straight up, carries it level over the container, and lowers it square onto the **container rim**.
- Press straight down on the lid edge nearest the front edge until it goes down onto the rim, then release, close on the far lid edge, and press straight down there the same way.

Hold the lid by its edge only. Never touch the **lid interior** and never press on the **lid top**.

The container is held for the whole press. The lid skirt only comes a short way down from the rim and the left gripper sits near the base, so the two never meet and the lid still seats all the way around. If the skirt lands on the gripper, the fix is to take the hold lower on the wall, not to let go of the container. The left gripper takes the load of the press, so the press is straight down and never at an angle, and the container does not slide, turn, or lift under it. Letting go to seat the lid is a violation, not a shortcut.

Check: the lid lies flat and square on the container rim with its edge down all the way around, no corner stands proud, no gap shows along any side, nothing is caught under the rim, the container is still flat and centered at the fill spot, and the left gripper is still in the container hold on its wall near the base.

Step 6: Stack the container

Goal: the lidded container squared on the stack at the stack spot.

- The **left gripper** is already in the **container hold** on the lidded container, so it takes no fresh grasp: from that same closed hold it lifts the container level off the fill spot and carries it level to the **stack spot**.
- The **right gripper** stays clear of the fill spot for the whole of this step. It does not steady the lidded container and does not travel with it.
- For the first container, lower it until its base is flat at the stack spot with the large compartment nearest the front edge, and release.
- For every container after it, lower it square onto the lid of the container below so their edges line up and the large compartment faces the same way, and release once it sits without rocking.

Keep the container level for the whole carry so nothing shifts between compartments.

There is no steady assist here. The left gripper never let go of the container, so nothing has to hold the lidded container still for a fresh grasp, and releasing it at the fill spot only to close on it again is a violation of the container hold. The right gripper's only contact with a container in the whole episode is the steady assist in the container row in Step 1, and it never goes anywhere near the stack.

Check: the container sits on the stack with its edges in line with the one below, it does not rock, its large compartment is nearest the front edge, the fill spot is clear, and the right gripper is empty and clear of the fill spot.

After the first, second, and third containers, return to Step 1 with the next container. After the fourth, go on to Step 7.

Step 7: Confirm the stack and end the episode

- Confirm the stack spot holds four lidded containers squared on each other, the fill spot is clear, the container row is empty, the scoop and the tongs are at their tool homes, and the lid rest is empty.
- Return both arms home, clear of the stack and the fill spot, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the stack apart, lift each lid off, and empty each compartment back into the bin its component came from. Never combine components.
2. Wash and dry the containers, the lids, the scoop, and the tongs.
3. Stand four clean dry containers open in the container row at the front left, each with its base flat and its large compartment nearest the front edge, and confirm the leftmost one is still inside the right arm's reach for the steady assist.
4. Stack four matching lids at the lid rest, inner face down, and set the scoop and the tongs in their own spots at the tool rest.
5. Top up the three bins so each holds more than one episode needs, and clear anything spilled off the table.
6. Wipe any debris off the table and confirm the surface is clean and dry. A wet table lets a container slide under a container hold, including under the lid press.
7. Confirm the fill spot and the stack spot are clear.
8. Replace a bin 1 component that has gone sticky enough that a level scoop will not hold its edge, bin 2 pieces too large to sit inside the left compartment, and bin 3 pieces stuck together.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** component 2 or component 3 goes into a container before component 1, component 3 goes in before component 2, a lid is seated before all three components are in, or a container leaves the fill spot before its lid is seated.
- **SOP rule broken:** Steps 1 to 6, the order never changes: container, component 1, component 2, component 3, lid, stack.
- **Coaching note:** bin 1, bin 2, bin 3, lid, stack. Four times over.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the scoop, the tongs, or a lid, or reaches a source bin or the lid rest; or the right gripper lifts or moves a container, or carries one to the stack spot. A steady assist in the container row in Step 1 is not this violation.
- **SOP rule broken:** Steps 1 to 6, the right gripper takes the tools, fills the compartments, and seats the lids, and the left gripper moves, holds, and stacks. The right gripper's only contact with a container is the steady assist in Step 1.
- **Coaching note:** each arm stays on its own side. The right gripper reaches a container in the row to hold it still for the left gripper's grasp and nothing else.

Violation: Steady assist used wrong

- **Visible cue:** the right gripper lifts, slides, turns, or rights a container, or carries it toward the fill spot or the stack spot; it is still on the container when the container leaves the table, or it travels with it; it holds a container at the fill spot, during any portioning, or while the lid is lowered or pressed; it touches a container on the stack; or it comes onto a container with the scoop, the tongs, or a lid in hand.
- **SOP rule broken:** Step 1, the steady assist is an empty right gripper holding an empty container still in the container row for the left gripper's grasp, released before the container

lifts, and the left gripper does every move of every container.

- **Coaching note:** the right gripper only holds it still, then lets go and backs off before the lift. One place only: the container row. Nothing at the fill spot needs steadying, because the left gripper is already holding it.

Violation: Component touched by a gripper

- **Visible cue:** a gripper closes on a component directly, or pushes, sweeps, or pats one with its fingers instead of moving it with the scoop or the tongs.
- **SOP rule broken:** Steps 2 to 4, component 1 moves by scoop and every piece of component 2 and component 3 moves by tongs.
- **Coaching note:** the tool touches the component. The gripper touches the tool.

Violation: Wrong tool, or two tools out at once

- **Visible cue:** the scoop lifts a piece of component 2 or component 3, the tongs move component 1, both the scoop and the tongs are out of the tool rest at the same time, or one tool is set on the table or left in a bin instead of going back to its tool home.
- **SOP rule broken:** Steps 2 to 4, the scoop moves component 1 and the tongs move component 2 and component 3, one tool at a time, each returned to its tool home before the other is taken.
- **Coaching note:** finish with one tool and put it back before you pick up the other.

Violation: Tool grasped wrong

- **Visible cue:** the right gripper closes on the scoop bowl instead of the middle of the handle, or holds the tongs at their tips or by one arm so closing the gripper does not close the tongs.
- **SOP rule broken:** Steps 2.1, 3, and 4, the scoop is held at the middle of its handle and the tongs across the middle of their arms.
- **Coaching note:** hold the handle, not the working end.

Violation: Wrong portion count

- **Visible cue:** a container gets other than two scoops of component 1, other than four pieces of component 2, or other than three pieces of component 3; a scoop is emptied into a compartment while still heaped above its edge; or the tongs take two pieces in one pick.
- **SOP rule broken:** Steps 2 to 4, two level scoops, four pieces, and three pieces per container, one piece per tongs pick.
- **Coaching note:** count as you go. Level the scoop in the bin, and take one piece at a time.

Violation: Component in the wrong compartment

- **Visible cue:** component 1 lands in the left compartment or the right compartment, a piece of component 2 lands in the large compartment or the right compartment, a piece of

component 3 lands in the large compartment or the left compartment, or a piece is left sitting on a divider or on the container rim.

- **SOP rule broken:** Steps 2 to 4, component 1 in the large compartment, component 2 in the left compartment, and component 3 in the right compartment.
- **Coaching note:** put the tool over the compartment before you tilt or open it.

Violation: Dropped piece put into a container

- **Visible cue:** a piece that has fallen on the table is picked up again and placed in a compartment, or spilled component 1 is scooped off the table and into a container or back into a bin.
- **SOP rule broken:** Steps 3 and 4, anything that has touched the table never goes into a container, and a fresh piece is taken from the bin instead.
- **Coaching note:** leave what falls where it fell and take a new one.

Violation: Container hold released or kept at the wrong time

- **Visible cue:** the left gripper is off the container while a scoop or a piece goes in, while the lid is lowered onto the rim, or while either lid edge is pressed; the left gripper releases at the fill spot after the lid is seated and closes on the container again for the carry; or the container slides, spins, or lifts off the table during any of it.
- **SOP rule broken:** Steps 1 to 6, the left gripper takes the container hold low on the wall the moment the container lands at the fill spot and keeps it unbroken through the portioning and the lid press, then lifts on that same grasp and opens only once the container is down on the stack. A right gripper hold never stands in for it.
- **Coaching note:** one grasp, low on the wall, all the way from the fill spot to the stack. The lid seats over the hold, so there is never a reason to let go.

Violation: Container set down wrong at the fill spot

- **Visible cue:** the container lands at the fill spot with its large compartment away from the front edge or turned sideways, with its base not flat, or off the fill spot.
- **SOP rule broken:** Step 1, the container sits flat and centered at the fill spot with the large compartment nearest the front edge.
- **Coaching note:** face it the same way every time. The large compartment is nearest you.

Violation: Lid seated wrong or lid interior touched

- **Visible cue:** at the end of Step 5 the lid is tilted, off center, standing proud at a corner, or showing a gap along a side; the lid is laid on without either press; a gripper contacts the lid interior or presses on the lid top; or a piece is caught under the rim.
- **SOP rule broken:** Step 5, the lid is held by its edge, lowered square onto the container rim, and pressed down at the near edge and then the far edge until it sits flat and square all the way around, with the left gripper holding the container throughout.

- **Coaching note:** square it first, then press near, then press far. Hold the skirt, never the inside.

Violation: Stack built wrong

- **Visible cue:** a container without a seated lid is set on the stack, a container is set beside the stack instead of on it, its edges do not line up with the container below, it rocks when released, its large compartment faces a different way from the rest, or the stack topples.
- **SOP rule broken:** Step 6, every container reaches the stack lidded and is lowered square on the one below with its edges in line and its large compartment nearest the front edge.
- **Coaching note:** lid first, then line up the edges before you open the gripper.

Violation: Bad pick, carry, or release

- **Visible cue:** two containers or two lids are lifted in one grasp, a thing stays in contact with the table while it moves between two zones, a gripper shifts or seats its hold again in the air, a container is carried tilted, or a gripper opens over the table or over open space instead of over the spot the thing is going to.
- **SOP rule broken:** Steps 1 to 6, one thing per pick, each lifted clear of the table before it moves, carried level without a new grip in the air, and released over its destination.
- **Coaching note:** one thing per pick, lift before you carry, set it down before you take a new hold, and know where it lands before you open the gripper.

Violation: Placement shoved or fiddled with

- **Visible cue:** a gripper nudges, pats, or slides a container or a lid into position after releasing it instead of lifting it clear and placing it again, or makes more than about two small corrections to settle a placement.
- **SOP rule broken:** Steps 1, 5, and 6, correct a bad placement by lifting the thing clear and placing it again, and each placement settles in one or two small corrections.
- **Coaching note:** lift it and lay it again rather than shoving it. If the first placement lands wide every time, tighten the approach rather than correcting after.

Violation: Dropped or knocked something over

- **Visible cue:** a container, a lid, or a tool falls to the table or the floor short of its spot, a source bin is knocked and shifts or tips, or the stack is knocked out of line.
- **SOP rule broken:** Steps 1 to 6, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with fewer than four containers stacked, a container still at the fill spot or in the container row, an unlidded container on the stack, a tool away from its

tool home, an arm away from home, or an arm still over the stack.

- **SOP rule broken:** Step 7, confirm the stack, the clear zones, and the tools; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective container:** cracked, warped, wet, or dirty, or a rim that will not take a lid under a correct press. Replace it before the next episode.
- **Defective lid:** cracked or warped, or one whose skirt will not go down over a correct rim under a correct press. Replace it before the next episode.
- **Defective tool:** a bent scoop, or tongs that do not close on a piece or do not spring open under a correct tongs grip. Replace it before the next episode.
- **Component out of spec:** a bin 1 component sticky enough that a level scoop will not hold its edge, a bin 2 piece too large to sit inside the left compartment, or bin 3 pieces stuck together so one pick lifts two. Replace the bin contents before the next episode.
- **Container slides or turns** under a correct container hold, including under a straight-down lid press, because the table is wet or slick. Wipe and dry the table before the next episode.
- **Stack spot out of reach:** the left arm cannot set the fourth container on the stack without extending to a joint limit. Move the stack spot in before the next episode.

Annotation subtasks (from SOP)

1. Move a container from the container row to the fill spot and take the container hold, with the right gripper steadying the container in the row where it is needed
2. Portion two level scoops of component 1 into the large compartment
3. Portion four pieces of component 2 into the left compartment
4. Portion three pieces of component 3 into the right compartment
5. Seat the lid on the container with the container hold kept on throughout
6. Lift the lidded container on that same hold, carry it to the stack spot, and square it on the stack
7. Repeat subtasks 1 to 6 for the second, third, and fourth containers
8. Confirm the stack and end the episode